Prothrombin complex concentrate, 4-factor, unactivated, from human plasma: Drug information - UpToDate

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Contributor Disclosures

For additional information see ”Prothrombin complex concentrate, 4-factor, unactivated, from human plasma: Patient drug information”

For abbreviations, symbols, and age group definitions show table

ALERT: US Boxed Warning

Arterial and venous thromboembolic complications:

Patients being treated with vitamin K antagonist therapy have underlying disease states that predispose them to thromboembolic events. Potential benefits of reversing vitamin K antagonists should be weighed against the potential risks of thromboembolic events, especially in patients with history of a thromboembolic event. Resumption of anticoagulation should be carefully considered as soon as the risk of thromboembolic events outweighs the risk of acute bleeding.

Both fatal and nonfatal arterial and venous thromboembolic complications have been reported with prothrombin complex concentrate in clinical trials and postmarketing surveillance. Monitor patients receiving prothrombin complex concentrate for signs and symptoms of thromboembolic events.

Prothrombin complex concentrate was not studied in subjects who had a thromboembolic event, myocardial infarction, disseminated intravascular coagulation, cerebral vascular accident, transient ischemic attack, unstable angina pectoris, or severe peripheral vascular disease within the prior 3 months. Prothrombin complex concentrate may not be suitable in patients with thromboembolic events in the prior 3 months.

Brand Names: US

- Balfaxar;
- Kcentra

Brand Names: Canada

- Beriplex P/N;
- Octaplex

Pharmacologic Category

- Blood Product Derivative;
- Hemostatic Agent;
- Prothrombin Complex Concentrate (PCC)

Dosing: Adult

Note: Prothrombin complex concentrate (human) [(factors II, VII, IX, X), protein C, protein S] contains therapeutic levels of factor VII component and should not be confused with factor IX complex (human) [factors II, IX, X] (Bebulin, Profilnine).

Life-threatening hemorrhage associated with direct factor Xa inhibitor or direct thrombin inhibitor

Life-threatening hemorrhage associated with direct factor Xa inhibitor or direct thrombin inhibitor (off-label use):

Note: Used for life-threatening bleeding or bleeding into a critical organ that is not controlled with maximal supportive measures (Ref). Dosage is expressed in units of factor IX activity and based on Kcentra studies.

Oral direct factor Xa inhibitor mediated (apixaban, betrixaban, edoxaban, rivaroxaban [if andexanet alfa unavailable]): **IV:** 2,000 units once **or** 25 to 50 units/kg once, in addition to other supportive measures as clinically indicated (eg, activated charcoal, antifibrinolytic agent) (Ref).

Direct thrombin inhibitor mediated (argatroban, dabigatran, bivalirudin [if idarucizumab or activated prothrombin complex concentrate unavailable]): **IV:** 50 units/kg in addition to other supportive measures as clinically indicated (eg, activated charcoal, hemodialysis, antifibrinolytic agent) (Ref).

Perioperative coagulopathy

Perioperative coagulopathy (off label use): Note: Limited data available; also refer to institutional protocols. Dosage is expressed in units of factor IX activity and based on Kcentra studies.

Cardiac surgery (cardiopulmonary bypass-induced coagulopathy): **IV:** 15 to 25 units/kg as a single dose (Ref). May also consider a single fixed dose (eg, >60 kg administer 2,000 units and ≤60 kg administer 1,500 units) (Ref); refer to institutional protocols.

Cardiac transplantation: Dosing based on preoperative INR (Ref):

INR <4: **IV:** 25 units/kg (maximum dose: 2,500 units).

INR 4 to 6: **IV:** 35 units/kg (maximum dose: 3,500 units).

INR >6: **IV:** 50 units/kg (maximum dose: 5,000 units).

Liver transplantation: **IV:** 25 to 40 units/kg as a single dose; higher dose guided by rotational thromboelastometry and/or laboratory values (Ref).

Coagulopathy following trauma**: IV:** 25 units/kg as a single dose (in combination with fresh frozen plasma) (Ref).

Reversal of oral direct factor Xa inhibitor in patients who require urgent procedure

Reversal of oral direct factor Xa inhibitor in patients who require urgent procedure (if andexanet alfa unavailable) (off-label use):

Note: Reversal agent should be administered only if the procedure cannot safely be performed while the patient is anticoagulated or cannot be delayed (Ref). Dosage is expressed in units of factor IX activity and based on Kcentra studies.

IV: 2,000 units once (Ref).

Reversal of vitamin K antagonist in patients with acute major bleeding or need for an urgent surgery/invasive procedure

Reversal of vitamin K antagonist in patients with acute major bleeding or need for an urgent surgery/invasive procedure:

Note: Individualize dosing based on severity of disorder, INR, extent and location of bleeding, and clinical status of patient. Repeat dosing with prothrombin complex concentrate is usually not necessary when given in combination with vitamin K. Dosage is expressed in units of factor IX activity. Balfaxar is only indicated for reversal of vitamin K antagonist in patients with need for an urgent surgery/invasive procedure. Kcentra, Beriplex P/N [Canadian product], and Octaplex [Canadian product] are indicated for reversal of vitamin K antagonist in patients with acute major bleeding **or** need for an urgent surgery/invasive procedure**.**

Balfaxar, Kcentra, Beriplex P/N [Canadian product]:

Weight-based dose:

Pretreatment INR: 2 to <4: **IV:** Administer 25 units/kg; maximum dose: 2,500 units.

Pretreatment INR: 4 to 6: **IV:** Administer 35 units/kg; maximum dose: 3,500 units.

Pretreatment INR: >6: **IV:** Administer 50 units/kg; maximum dose: 5,000 units.

Fixed dose: **Note:** Dosing based on Kcentra studies.

IV: Initial: 1,000 to 2,000 units once; for intracranial hemorrhage use, 1,500 to 2,000 units is recommended; repeat dosing has not been adequately studied and is not typically recommended unless INR reversal is inadequate (Ref).

Octaplex [Canadian product]:

Pretreatment INR: 2 to 2.5: **IV:** 22.5 to 32.5 units/kg; maximum dose: 3,000 units (or 120 mL).

Pretreatment INR: 2.5 to 3: **IV:** 32.5 to 40 units/kg; maximum dose: 3,000 units (or 120 mL).

Pretreatment INR: 3 to 3.5: **IV:** 40 to 47.5 units/kg; maximum dose: 3,000 units (or 120 mL).

Pretreatment INR: >3.5: **IV:** >47.5 units/kg; maximum dose: 3,000 units (or 120 mL).

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. However, need for dosage adjustment unlikely as limited excretion in the urine (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1, 2, or 3 obesity (BMI ≥30 kg/m^2): IV: For patients ≤100 kg, use **actual body weight** for weight-based dosing (Ref). For patients >100 kg, use a maximum 100 **kg** for dosing weight. Refer to adult dosing for indication-specific doses.

Rationale for recommendations: There is a lack of studies evaluating the influence of obesity on prothrombin complex concentrate (human) dosing or pharmacokinetics. The pharmacokinetics in healthy subjects without obesity is complicated with different included clotting factors having vastly different half-lives. Observational data support use of actual body weight (with a dose cap strategy) as the preferred weight metric in patients with a BMI ≥30 kg/m^2 (Ref). A retrospective, multicenter study evaluated ≤100 kg and >100 kg cohorts in patients receiving warfarin who received prothrombin complex concentrate for elevated INR and life-threatening bleed or required emergent surgery. This study did not show any difference in normalization of INR or thrombosis (Ref). Some retrospective studies have evaluated fixed dosing for vitamin K antagonist reversal; however, there are limited data describing fixed-dosing strategies in patients with obesity, particularly in patients with BMI >40 kg/m^2 or high-risk populations such as intracranial hemorrhage. Further studies are required to evaluate weight metrics and clinical outcomes in patients with obesity, especially in patients with BMI >40 kg/m^2.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

Octaplex [Canadian product]: Adolescents ≥17 years: Refer to adult dosing.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer’s labeling.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer’s labeling.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Adverse reactions reported in adults.

>10%: Nervous system: Asthenia (12%)

1% to 10%:

Cardiovascular: Arteriovenous fistula site complication (clot: 1%), atrial fibrillation (4%), deep vein thrombosis (1%), heart failure (≤3%), hypotension (7%; including hemorrhagic shock, orthostatic hypotension), tachycardia (5%), thrombosis (microthrombosis of toes: 1%), venous thrombosis (calf: 1%; radial vein: 1%), unstable angina pectoris (1%)

Endocrine & metabolic: Hypervolemia (≤3%), hypokalemia (5%)

Gastrointestinal: Abdominal pain (3%), diarrhea (2%), nausea and vomiting (6%)

Genitourinary: Dysuria (5%)

Hematologic and oncologic: Anemia (6%), bruise (≤4%), subcutaneous hematoma (≤4%)

Nervous system: Cerebrovascular accident (1% to 2%), headache (7%), insomnia (5%)

Respiratory: Dyspnea (≤4%), hypoxia (≤4%), pleural effusion (4%), pulmonary edema (2%), respiratory distress (≤4%)

Miscellaneous: Laceration (≤4%)

Frequency not defined:

Cardiovascular: Peripheral venous insufficiency

Hematologic & oncologic: Hemorrhage (including subdural hemorrhage)

Postmarketing:

Cardiovascular: Bradycardia, circulatory shock, hypertension, thromboembolic complications (including acute myocardial infarction, arterial thrombosis, pulmonary embolism)

Dermatologic: Pruritus, skin rash, urticaria

Hematologic & oncologic: Disseminated intravascular coagulation

Hypersensitivity: Hypersensitivity reaction (including anaphylactic shock, anaphylaxis, angioedema, and nonimmune anaphylaxis)

Nervous system: Chills, paresthesia, tremor

Respiratory: Respiratory failure

Miscellaneous: Fever

Contraindications

Balfaxar: Hypersensitivity (ie, anaphylaxis or severe systemic reaction) to prothrombin complex concentrate (PCC) or any component of the formulation; known allergy to heparin or history of heparin-induced thrombocytopenia (product contains heparin), immunoglobulin A (IgA) deficiency, with known antibodies against IgA.

Kcentra, Beriplex P/N [Canadian product]: Hypersensitivity (ie, anaphylaxis or severe systemic reaction) to PCC or any component of the formulation including heparin, factors II, VII, IX, X, protein C and S, antithrombin III and human albumin; disseminated intravascular coagulation (DIC); known heparin-induced thrombocytopenia (product contains heparin).

Octaplex [Canadian product]: Hypersensitivity to PCC or any component of the formulation; heparin-induced thrombocytopenia type II or known allergy to heparin (product contains heparin); non-life-threatening bleeding episodes in individuals with recent myocardial infarction, high risk of thrombosis, or angina pectoris; non-life-threatening bleeding episodes in individuals with untreated disseminated intravascular coagulation (DIC) who can be given fresh frozen plasma (FFP); coagulation disorders due to chronic liver disease or liver transplantation; bleeding associated with hepatic parenchymal disorders, esophageal varices, or major hepatic surgery; immunoglobulin A (IgA) deficiency, with known antibodies against IgA

Warnings/Precautions

Concerns related to adverse effects:

• Hypersensitivity reactions: Hypersensitivity reaction (eg, angioedema, bronchospasm, dyspnea, flushing, hypotension, nausea/vomiting, pulmonary edema, urticaria, tachycardia, tachypnea) may occur; if serious reaction occurs, discontinue administration and begin appropriate treatment.

Disease-related concerns:

• Hypercoagulopathy: Administration of prothrombin complex concentrate (PCC) may exacerbate underlying hypercoagulable states in recipients of vitamin K antagonists.

Dosage form specific issues:

• Heparin: Formulations may contain heparin.

• Human plasma: Product of human plasma; may potentially contain infectious agents which could transmit disease. Screening of donors, as well as testing and/or inactivation or removal of certain viruses, reduces the risk. Infections thought to be transmitted by this product should be reported to the manufacturer.

Other warnings/precautions:

• Appropriate use: **PCC (Human) [(Factors II, VII, IX, X), Protein C, Protein S] contains therapeutic levels of factor VII component** and should not be confused with Factor IX complex (Human) [Factors II, IX, X] (Bebulin, Profilnine) which contains low or nontherapeutic levels of factor VII.

• Coagulation factor deficiency: Hepatic synthesis of the prothrombin complex (Factors II, VII, IX and X) coagulation factors is vitamin K dependent. Severe hepatic dysfunction, inadequate absorption of vitamin K (eg, pancreatic disorders, diarrhea) or vitamin K antagonist therapy or overdose may lead to coagulation factor deficiencies. In patients with an acquired deficiency of the vitamin K dependent coagulation factors, administer PCC only if a rapid correction (eg, emergency surgery, major bleeding) is necessary. If not indicated and caused by Vitamin K antagonist therapy, coagulation factor deficiencies may be managed by reducing or discontinuing therapy of the vitamin K antagonist and/or administration of vitamin K.

Dosage Forms Considerations

Note: Potency on the vial label is expressed in terms of the Factor IX nominal strength (500 units or 1,000 units). Consult individual vial labels for exact potency within each vial.

Balfaxar:

Total active ingredient composition of 500 unit vial:

Factor II: 340 to 500 units

Factor VII: 240 to 400 units

Factor IX: 400 to 640 units

Factor X: 300 to 540 units

Protein C: 320 to 560 units

Protein S: 240 to 600 units

Heparin: 80 to 384 units

Total active ingredient composition of 1,000 unit vial:

Factor II: 680 to 1,000 units

Factor VII: 480 to 800 units

Factor IX: 800 to 1,280 units

Factor X: 600 to 1,080 units

Protein C: 640 to 1,120 units

Protein S: 480 to 1,200 units

Heparin: 160 to 768 units

Kcentra, Beriplex P/N [Canadian product]:

Total active ingredient composition of 500 unit vial:

Factor II: 380 to 800 units

Factor VII: 200 to 500 units

Factor IX: 400 to 620 units (Kcentra) or 500 units (Beriplex P/N)

Factor X: 500 to 1,020 units

Protein C: 420 to 820 units

Protein S: 240 to 680 units

Heparin: 8 to 40 units (Kcentra) or up to 40 units (Beriplex P/N)

Total active ingredient composition of 1,000 unit vial:

Factor II: 760 to 1,600 units

Factor VII: 400 to 1,000 units

Factor IX: 800 to 1,240 units (Kcentra) or 1,000 units (Beriplex P/N)

Factor X: 1,000 to 2,040 units

Protein C: 840 to 1,640 units

Protein S: 480 to 1,360 units

Heparin: 16 to 80 units (Kcentra) or up to 80 units (Beriplex P/N)

Octaplex [Canadian product]: **Note:** Potency on the vial label is expressed in terms of the Factor IX nominal strength (500 units or 1,000 units). Consult individual vial labels for exact potency within each vial.

Total active ingredient composition of 500 unit vial:

Factor II: 280 to 760 units

Factor VII: 180 to 480 units

Factor IX: 500 units

Factor X: 360 to 600 units

Protein C: 260 to 620 units

Proteins S: 240 to 640 units

Heparin: 80 to 310 units

Total active ingredient composition of 1,000 unit vial:

Factor II: 560 to 1,520 units

Factor VII: 360 to 960 units

Factor IX: 1,000 units

Factor X: 720 to 1,200 units

Protein C: 520 to 1,240 units

Protein S: 480 to 1,280 units

Heparin: 160 to 620 units

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Kit, Intravenous [preservative free]:

Kcentra: ~500 units, ~1000 units [pyrogen free; contains albumin human, heparin - flush/excipient (porcine)]

Solution Reconstituted, Intravenous:

Balfaxar: 500 units (1 ea); 1000 units (1 ea) [contains heparin - flush/excipient (porcine)]

Generic Equivalent Available: US

No

Pricing: US

Kit (Kcentra Intravenous)

500 unit (Price provided is per AHF Unit): $3.58

1000 unit (Price provided is per AHF Unit): $3.58

Solution (reconstituted) (Balfaxar Intravenous)

500 unit (Price provided is per AHF Unit): $3.78

1000 unit (Price provided is per AHF Unit): $3.78

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution Reconstituted, Intravenous:

Beriplex P/N: 500 units (1 ea); 1000 units (1 ea) [contains albumin human, heparin - flush/excipient (porcine)]

Octaplex: 500 units (1 ea); 1000 units (1 ea) [contains heparin - flush/excipient (porcine), polysorbate 80]

Administration: Adult

Balfaxar, Kcentra, Beriplex P/N [Canadian product]: **IV:** Administer at room temperature at a rate of 0.12 mL/kg/minute (~3 **units**/kg/minute); do **not** exceed 8.4 mL/minute (~210 **units**/minute). Do not allow blood to enter into syringe (fibrin clot formation may occur).

Octaplex [Canadian product]: **IV:** Administer at a rate of 1 mL/minute initially, followed by 2 to 3 mL/minute.

Preparation for Administration: Adult

Reconstitute with 20 mL (500 unit vial) or 40 mL (1,000 unit vial) of provided diluent (SWFI). Prior to reconstitution, allow diluent (SWFI) and prothrombin complex concentrate (PCC) vials to warm to room temperature. Refer to manufacturer’s labeling for detailed preparation instructions.

Note: Potency on the vial label is expressed in terms of the factor IX nominal strength (500 units or 1,000 units). Consult individual vial labels for exact potency within each vial. Manufacturer’s labeling recommends the exact amount of factor IX units in each vial be used when calculating and preparing the total dose to be administered. However, the Institute for Safe Medication Practices recommends using the nominal strength as described in clinical trials (Ref).

Administration: Pediatric

IV: Octaplex [Canadian product]: Administer at a rate of 1 mL/minute initially, followed by 2 to 3 mL/minute. Reduce infusion rate or interrupt infusion if patient’s pulse rate increases significantly. Do not mix with other drugs; if administering through IV line, ideally a new, clean line should be used; if an existing line must be used, flush with NS or D5W.

Preparation for Administration: Pediatric

IV: Octaplex [Canadian product]: Use the diluent (SWFI) and transfer device provided. Prior to reconstitution, allow diluent (SWFI) and prothrombin complex concentrate (PCC) vials to warm to room temperature. Aseptically push the plastic spike at the blue end of the Mix2Vial transfer set through the center of the stopper of the diluent vial. After carefully removing only the clear package from the Mix2Vial transfer set, invert the diluent vial with the transfer set still attached and push the plastic spike through the center of the stopper of PCC vial; diluent will automatically transfer. While still attached, gently swirl PCC vial to ensure product is dissolved; do not shake. Disconnect the diluent vial and the blue part of the Mix2Vial transfer set; contents of PCC vial are now available for removal by screwing a syringe onto the transfer set. Invert vial and withdraw amount needed. Remove syringe from transfer set and attach an administration set to the syringe. Reconstituted solution should be a colorless to slightly blue solution. Do not use if particulate matter or discoloration is observed; a blue color is not interpreted as discoloration. See prescribing information for detailed reconstitution instructions.

Storage/Stability

Balfaxar: Store intact vials at 2°C to 25°C (36°F to 77°F). Do not freeze. Store vials in the original package to protect from light. Reconstituted solution should be administered immediately, but may be stored for up to 8 hours at 20°C to 25°C (68°F to 77°F).

Kcentra, Beriplex P/N [Canadian product]: Store intact vials at 2°C to 25°C (36°F to 77°F); do not freeze. Store vials in the original carton to protect from light. Reconstituted product may be stored at 2°C to 25°C (36°F to 77°F) and used within 4 hours (Kcentra) or 3 hours (Beriplex P/N) following reconstitution. If cooled, warm to 20°C to 25°C (68°F to 77°F) prior to administration.

Octaplex [Canadian product]: Store at 2°C to 25°C (36°F to 77°F); do not freeze. Store vials in the original carton to protect from light. Reconstituted solution should be administered immediately, but may be stored for up to 8 hours at 2°C to 25°C (36°F to 77°F) if sterility is maintained.

Use: Labeled Indications

Reversal of vitamin K antagonist in patients with acute major bleeding or need for an urgent surgery/invasive procedure: Urgent reversal of acquired coagulation factor deficiency induced by vitamin K antagonist (eg, warfarin) therapy in adult patients with acute major bleeding (Kcentra, Beriplex P/N [Canadian product], Octaplex [Canadian product]) or with a need for an urgent surgery/invasive procedure (Balfaxar, Kcentra, Beriplex P/N [Canadian product], Octaplex [Canadian product]).

Use: Off-Label: Adult

Life-threatening bleeding associated with direct factor Xa inhibitor or direct thrombin inhibitor; Perioperative coagulopathy; Reversal of oral direct factor Xa inhibitor in patients who require urgent procedure (if andexanet alfa unavailable)

Medication Safety Issues

Sound-alike/look-alike issues:

The term "Prothrombin Complex Concentrate" or "PCC" has been used to describe both Factor IX Complex (Human) [Factors II, IX, X] and Prothrombin Complex Concentrate (Human) [(Factors II, VII, IX, X), Protein C, Protein S]. **Prothrombin Complex Concentrate (Human) [(Factors II, VII, IX, X), Protein C, Protein S] (Balfaxar, Beriplex P/N [Canadian product], Kcentra, Octaplex [Canadian product]) contains therapeutic levels of factor VII component** and should not be confused with Factor IX complex (Human) [Factors II, IX, X] (Bebulin, Profilnine) which contains low or nontherapeutic levels of factor VII.

Other safety concerns:

Potency on the vial label is expressed in terms of the factor IX nominal strength (500 units or 1,000 units). Consult individual vial labels for exact potency within each vial. Manufacturer's labeling recommends the exact amount of factor IX units in each vial be used when calculating and preparing the total dose to be administered. However, the Institute for Safe Medication Practices recommends using the nominal strength as described in clinical trials (ISMP 2017).

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Antifibrinolytic Agents: May increase adverse/toxic effects of Prothrombin Complex Concentrate (Human) [(Factors II, VII, IX, X), Protein C, and Protein S]. Specifically, the risk for thrombosis may be increased. *Risk X: Avoid*

Pregnancy Considerations

Animal reproduction studies have not been conducted. This product is derived from purified human plasma.

Breastfeeding Considerations

It is not known if prothrombin complex concentrate is present in breast milk.

The manufacturer recommends that prothrombin complex concentrate be administered only if clearly needed when treating a breastfeeding patient.

Monitoring Parameters

INR (baseline and at 30 minutes post dose); clinical response during and after treatment; signs of thromboembolism

Mechanism of Action

Prothrombin complex concentrate provides an increase in the levels of the vitamin K-dependent coagulation factors (II, VII, IX, and X) with the addition of protein C and protein S. Coagulation factors II, IX, and X are part of the intrinsic coagulation pathway, while factor VII is part of the extrinsic coagulation pathway. In the *extrinsic* pathway, damaged blood vessels release endothelial tissue factor (TF) which complexes with factor VII to form TF-factor VIIa. Within the *intrinsic* pathway, factor IX is converted to IXa. Factor IXa (as well as TF-factor VIIa) converts factor X to factor Xa in the final common pathway of coagulation. Factor Xa activates prothrombin (factor II) into thrombin (IIa) which converts fibrinogen into fibrin resulting in clot formation. Proteins C and S are vitamin K-dependent inhibiting enzymes involved in regulating the coagulation process. Protein S serves as a cofactor for protein C which is converted to activated protein C (APC). APC is a serine protease which inactivates factors Va and VIIIa, limiting thrombotic formation.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Rapid; significant INR decline within 30 minutes.

Duration: INR reversal usually maintained for up to ≥24 hours.

Distribution: V_{dss} (Kcentra, Beriplex P/N [Canadian product]): Factor II: 71.4 mL/kg; Factor VII: 45 mL/kg; Factor IX: 114.3 mL/kg; Factor X: 55.5 mL/kg; Protein C: 62.2 mL/kg; Protein S: 78.8 mL/kg. **Note:** Distribution data for Balfaxar and Octaplex (Canadian product) not reported in the manufacturer's labeling.

Half-life elimination: Factor II: 48 to 60 hours; Factor VII: 1.5 to 6 hours; Factor IX: 20 to 24 hours; Factor X: 24 to 48 hours; Protein C: 1.5 to 6 hours; Protein S: 24 to 48 hours.

Note: Half-lives may be significantly reduced in severe hepatocellular damage, DIC, or extended catabolic metabolism.

Patient Counseling Points

(For additional information see "Prothrombin complex concentrate, 4-factor, unactivated, from human plasma: Patient drug information")

What is this drug used for?

• It is used to undo the effects of certain blood thinners like warfarin.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Headache

• Upset stomach or throwing up

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• Flushing

• A fast heartbeat

• Anxiety

• Severe dizziness or passing out

• Fast breathing

• Pain when passing urine

• Weakness on 1 side of the body, trouble speaking or thinking, change in balance, drooping on one side of the face, or blurred eyesight

• Feeling very tired or weak

• Blood clot like chest pain or pressure; coughing up blood; shortness of breath; swelling, warmth, numbness, change of color, or pain in a leg or arm; or trouble speaking or swallowing

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

Find brand name(s) by country or region (for United States and Canada see separate Brand Names sections)

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Beriplex p/n;
- (AR) Argentina: Beriplex p/n | Octaplex | Protromplex total;
- (AT) Austria: Octaplex;
- (BE) Belgium: Octaplex;
- (BG) Bulgaria: Octaplex;
- (BR) Brazil: Beriplex p/n | Octaplex;
- (CH) Switzerland: Octaplex 500;
- (CL) Chile: Beriplex p/n | Octaplex;
- (CO) Colombia: Beriplex | Hyfact | Octaplex;
- (CR) Costa Rica: Octaplex;
- (CZ) Czech Republic: Beriplex | Ocplex;
- (DE) Germany: Octaplex | PPSB human SD/Nano;
- (EC) Ecuador: Octaplex;
- (EE) Estonia: Octaplex;
- (ES) Spain: Beriplex;
- (FI) Finland: Confidex | Octaplex;
- (FR) France: Confidex | Octaplex;
- (GB) United Kingdom: Octaplex;
- (GR) Greece: Beriplex | Octaplex;
- (HK) Hong Kong: Beriplex p/n;
- (HU) Hungary: Beriplex p/n | Prothrombin complex;
- (ID) Indonesia: Octaplex;

- (IE) Ireland: Octaplex;
- (IT) Italy: Confidex | Pronativ;
- (JP) Japan: Kcentra;
- (LB) Lebanon: Beriplex;
- (LT) Lithuania: Octaplex;
- (LV) Latvia: Octaplex;
- (MX) Mexico: Confidex | Octapro;
- (MY) Malaysia: Octaplex;
- (NL) Netherlands: Octaplex;
- (NO) Norway: Confidex | Octaplex;
- (PE) Peru: Octaplex;
- (PL) Poland: Octaplex;
- (PT) Portugal: Beriplex | Octaplex;
- (PY) Paraguay: Beriplex p/n | Octaplex;
- (QA) Qatar: Octaplex | Prothromplex Total;
- (RU) Russian Federation: Coaplex | Octaplex;
- (SA) Saudi Arabia: Beriplex p/n | Octaplex;
- (SE) Sweden: Confidex;
- (SG) Singapore: Octaplex;
- (SI) Slovenia: Beriplex;
- (SK) Slovakia: Beriplex | Octaplex;
- (TN) Tunisia: Octaplex;
- (UA) Ukraine: Octaplex;
- (UY) Uruguay: Octaplex;
- (ZA) South Africa: Octaplex

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